Civil Engineering

Trigonometry, Bearings and Complex Numbers

Revision Booklet

Use these questions for CAT, assignment preparation and summative revision.

Formula Recall

1. State where this formula is applied and create one original example.

$$a^2+b^2=c^2$$

2. State where this formula is applied and create one original example.

$$\sin\theta=\frac{O}{H}$$

3. State where this formula is applied and create one original example.

$$c^2=a^2+b^2-2ab\cos C$$

4. State where this formula is applied and create one original example.

$$\frac{a}{\sin A}=\frac{b}{\sin B}=\frac{c}{\sin C}$$

5. State where this formula is applied and create one original example.

$$z=a+bi$$

Structured Revision

1. A realistic civil engineering task involves slopes. Formulate the calculation, solve it, and interpret your result.

2. A realistic civil engineering task involves triangulation. Formulate the calculation, solve it, and interpret your result.

3. A realistic civil engineering task involves bearings. Formulate the calculation, solve it, and interpret your result.

4. A realistic civil engineering task involves polar representation. Formulate the calculation, solve it, and interpret your result.

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References and Learning Sources

These learner materials are original notes prepared for Mr Mwirigi Mathematics Training Hub. The topic sequencing and level are informed by standard engineering mathematics references and the uploaded curriculum/OS materials. No textbook pages or solution-manual pages have been reproduced.

1. John Bird, Engineering Mathematics, Newnes / Elsevier.

2. K. A. Stroud and Dexter J. Booth, Engineering Mathematics, Palgrave Macmillan.

3. Uploaded KNEC solved mathematics materials for examination-style orientation.

4. Civil Engineering, Building Technology and Land Survey curriculum and occupational-standard materials used for local alignment.